Samples

TerraLID ID

ID and name: S0 terralid_sample_id
Provided by: TerraLID system
Obligation: mandatory
Occurrences: 1
Definition: The ID of the sample in the TerraLID database.
Allowed values and other constraints: t.b.d.
Example: t.b.d.

1. Sample identifiers

ID and name: S1 sample_identifiers
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Identifiers assigned to the sample.

with the two subproperties:

1.1 Laboratory ID

ID and name: S1.1 sample_id_lab
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The identifier by which the sample was identified in the lab or publication.
Allowed values and other constraints: free text
Example: 2024/02

1.2 Persistent Identifier

ID and name: S1.2 sample_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Persistent identifier(s) assigned to the analysed material.

with the two subproperties:

1.2.1 Value of persistent Identifier

ID and name: S1.2.1 sample_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: A persistent identifier assigned to the analysed material.
Allowed values and other constraints: valid persistent identifier according to `S1.2.2 Type of persistent identifier`
Example: 10.60510/ICDP5054ESYI201

1.2.2 Type of persistent identifier

ID and name: S1.2.2 sample_pid_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The type of a persistent identifier that was assigned to the sample.
Allowed values and other constraints: controlled vocabulary

2. Objective of sampling

ID and name: S2 sample_objective
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Short note for which purpose the sample was originally taken.
Allowed values and other constraints: free text
Example: provenance analysis

3. Sampled material

ID and name: S3 sample_material
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The material that was sampled. This may often be the material recorded at `O12 Material` but can also be different, especially when a heterogeneous material is sampled.
Allowed values and other constraints: controlled vocabulary

4. Sampling location

ID and name: S4 sample_location
Provided by: data provider
Obligation: recommended

Occurrences: 0–1
Definition: Information about where on the object the sample was taken.

with the two subproperties:

4.1 Description of sampling location

ID and name: S4.1 sample_location_description
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Description of location the sample was taken from.
Allowed values and other constraints: free text
Example: edge of the coin

4.2 Photo of sampling location

ID and name: S4.2 sample_location_photo
Provided by: data provider
Obligation: optional
Occurrences: 0–1
Definition: Photograph of the sample location.
Allowed values and other constraints: Image file (File size must be smaller than 2 MB)

5. Sample type

ID and name: S5 sample_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of the sampled material.
Allowed values and other constraints: controlled vocabulary

6. Sample weight

ID and name: S6 sample_weight
Provided by: data provider
Obligation: optional
Occurrences: 0–1
Definition: The weight of the sample before analysis.

with the two subproperties:

6.1 Value

ID and name: S6.1 sample_weight_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The value of the weight.
Allowed values and other constraints: decimal number
Example: 3.25

6.2 Unit

ID and name: S6.2 sample_weight_unit
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: SI unit in which the weight is given.
Allowed values and other constraints: controlled vocabulary

7. Sampling method

ID and name: S7 sample_method
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The method used to take the sample.
Allowed values and other constraints: controlled vocabulary

8. Sample condition

ID and name: S8 sample_condition
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The state of the sample after analysis.
Allowed values and other constraints: controlled vocabulary

9. Sampling date

ID and name: S9 sample_date
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Date when the sample was taken.

Allowed values and other constraints: date formatted as YYYY-MM-DD
Example: 2024-02-24

10. Sampling laboratory

ID and name: S10 sample_laboratory
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Name (and address) of the laboratory, in which the sample was taken.
Allowed values and other constraints: free text
Example: Geochemistry laboratory of the University of Dreamland

11. Sample description

ID and name: S11 sample_description
Provided by: data provider
Obligation: optional
Occurrences: 0–1
Definition: Additional information about the sampling process not captured elsewhere.
Allowed values and other constraints: free text
Example: Sample was drilled with diamond-sputtered steel drill.

12. Bulk Pb concentration

ID and name: S12 sample_chemistry_pb
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: The bulk lead concentration of the sample.

with the eight subproperties:

12.1 Analytical method

ID and name: B4.1 chemistry_method
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The method used to determine the chemical composition.
Allowed values and other constraints: controlled vocabulary

12.2 Analysed compound

ID and name: B4.2 chemistry_compound
Provided by: data provider

Obligation: mandatory
Occurrences: 1–n
Definition: The analysed chemical compound (chemical element, oxide, isotope).
Allowed values and other constraints: controlled vocabulary, the actual measured compound must be reported if available (e.g., the isotope for mass spectrometric values rather than the element).

12.3 Value

ID and name: B4.3 chemistry_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The concentration of the analysed chemical compound.

with the two subproperties:

12.3.1 ABSOLUTE VALUE

ID and name: B4.3.1 chemistry_value_absolute
Provided by: data provider
Obligation: recommended
Occurrences: 1–n
Definition: The concentration of the analysed chemical compound.
Allowed values and other constraints: decimal number
Example: 15.3

12.3.2 QUALITATIVE VALUE

ID and name: B4.3.2 chemistry_value_relative
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Value of qualitative results and other non-numeric values such as the reporting of detection or quantification limits rather than values.
Allowed values and other constraints: free text
Example: +++, tr, <0.4, bdl = 0.5 wt%

12.4 Unit

ID and name: B4.4 chemistry_unit
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The unit in which the concentration of the analysed chemical compound is given.
Allowed values and other constraints: controlled vocabulary

12.5 Uncertainty type

ID and name: B4.5 chemistry_uncertainty_type
Provided by: data provider

Obligation: recommended
Occurrences: 0–n
Definition: The type of analytical uncertainty.
Allowed values and other constraints: controlled vocabulary

12.6 Confidence level

ID and name: B4.6 chemistry_uncertainty_sigma
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Multiple of standard deviation (sigma) of the reported absolute analytical uncertainty.
Allowed values and other constraints: 1, 2, 3
Example: 2

12.7 Uncertainty value

ID and name: B4.7 chemistry_uncertainty_value
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Value of the absolute analytical uncertainty.
Allowed values and other constraints: decimal number
Example: 0.3

13. Sampling person

ID and name: S13 sample_creator
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Information about the person that took the sample.

with the nine subproperties:

13.1 Role

ID and name: B1.1 person_role
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The role in which the person is related to the linked information.
Allowed values and other constraints: controlled vocabulary
Example: Author

13.2 First names

ID and name: B1.2 person_name_first
Provided by: data provider, API (ORCID ID)
Obligation: recommended
Occurrences: 0–1
Definition: The first and middle name(s) of the person.
Allowed values and other constraints: free text
Example: Jane

13.3 Last Name

ID and name: B1.3 person_name_last
Provided by: data provider, API (ORCID ID)
Obligation: mandatory
Occurrences: 1
Definition: The last name of the person.
Allowed values and other constraints: free text
Example: Doe

13.4 Persistent Identifier

ID and name: B1.4 person_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: Persistent identifier(s) assigned to the person.

with the two subproperties:

13.4.1 VALUE OF PERSISTENT IDENTIFIER

ID and name: B1.4.1 person_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: A persistent identifier assigned to the analysed material.
Allowed values and other constraints: Valid persistent identifier according to B1.4.2 Type of persistent identifier
Example: 0000–0001–2345–678X

13.4.2 TYPE OF PERSISTENT IDENTIFIER

ID and name: B1.4.2 person_pid_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of the persistent identifier.
Allowed values and other constraints: controlled vocabulary

13.5 Affiliation name

ID and name: B1.5 person_affiliation_name
Provided by: data provider, API (ORCID ID, ROR ID)
Obligation: mandatory
Occurrences: 1–n
Definition: The name of the person's affiliation.
Allowed values and other constraints: free text
Example: Institute of Time Travels

13.6 ROR ID

ID and name: B1.6 person_affiliation_ror
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The ROR of the person's affiliation.
Allowed values and other constraints: is valid ROR ID
Example: 09af7gtg53

13.7 Address

ID and name: B1.7 person_affiliation_address
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The address of the person's affiliation.
Allowed values and other constraints: free text
Example: Teleporter Avenue 123, Ankh–Morpok, United States of Humanities

13.8 Mail address

ID and name: B1.8 person_mail
Provided by: data provider, API (ORCID ID)
Obligation: recommended
Occurrences: 0–n
Definition: The mail address of the person.
Allowed values and other constraints: is valid mail address
Example: jane.doe@timetravels.int

13.9 Website

ID and name: B1.9 person_url
Provided by: data provider, API (ORCID ID)
Obligation: optional
Occurrences: 0–1
Definition: The URL of a person.
Allowed values and other constraints: is valid URL
Example: https://www.timetravels.int/members/jane–doe

14. Sample status

ID and name: S14 sample_status
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Information about the current status of the sample and how to access it.

with the two subproperties:

14.1 Institution

ID and name: B2.1 status_institution
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: The institution at which the material is located.

with the five subproperties:

14.1.1 NAME

ID and name: B2.1.1 status_institution_name
Provided by: data provider, API (ROR ID)
Obligation: mandatory
Occurrences: 1
Definition: Name of the institution.
Allowed values and other constraints: free text
Example: Institute of Time Travels

14.1.2 ROR

ID and name: B2.1.2 status_institution_ror
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: ROR of the institution.
Allowed values and other constraints: is valid ROR ID
Example: 09af7gtg53

14.1.3 ADDRESS

ID and name: B2.1.3 status_institution_address
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Address of the institution.

Allowed values and other constraints: free text
Example: Teleporter Avenue 123, Ankh–Morpok, United States of Humanities

14.1.4 STORAGE LOCATION

ID and name: B2.1.4 status_institution_location
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Location of the item within the institution.
Allowed values and other constraints: free text
Example: Archive 9, shelf 3, box 1

14.1.5 CONTACT

ID and name: B2.1.5 status_institution_contact
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Contact information for inquiries about the material. This may include, for example, a mail address or phone number of the respective department within the institution or the identification of a specific contact person as well as constraints on the availability of the point of contact such as opening hours.
Allowed values and other constraints: free text
Example: t.b.d.

14.2 Accessibility

ID and name: B2.2 status_accessibility
Provided by: data provider
Obligation: recommended
Occurrences: 0–1
Definition: Is the material accessible to other researchers and do restrictions apply?
Allowed values and other constraints: controlled vocabulary

15. Relations

ID and name: S15 sample_relation
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Information about related entities, including analysis made on the sample, and other research output such as publications providing relevant information about the sample. This includes any literature from which information about the sample was extracted.

with the five subproperties:

15.1 Persistent Identifier

ID and name: B5.1 relation_pid
Provided by: data provider
Obligation: recommended
Occurrences: 0–n
Definition: The persistent identifier or TerraLID ID associated with a resource or related research output. Mandatory if referring to another entity in the TerraLID database; in this case, the TerraLID ID must be used and `B5.1.2 Type` is "TerraLID". Within a submission, the identifiers of the submitted records may be used; the TerraLID system replaces them with the assigned TerraLID IDs.

with the two subproperties:

15.1.1 VALUE

ID and name: B5.1.1 relation_pid_value
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: Value of the persistent identifier or TerraLID ID. **Allowed values and other constraints:** valid persistent identifier according to the options listed in `B5.1.2 Type`, or a TerraLID ID if `B5.1.2 Type` = "TerraLID". **Example:** 10.60510/ICDP5054ESYI201

15.1.2 TYPE

ID and name: B5.1.2 relation_pid_type
Provided by: data provider
Obligation: mandatory
Occurrences: 1
Definition: The type of the identifier.
Allowed values and other constraints: controlled vocabulary (identifier). Use "TerraLID" for entities in the TerraLID database.

15.2 Full reference

ID and name: B5.2 relation_text
Provided by: data provider, API (DOI)
Obligation: recommended
Occurrences: 0–1
Definition: The full reference of a publication.
Allowed values and other constraints: free text
Example: Palinkaš, L. A., 1985, Lead isotope patterns in galenas from some selected ore deposits in Croatia and NW Bosnia, Geološki Vjesnik, 38, 175–89.

15.3 Kind of relation

ID and name: B5.3 relation_kind
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n

Definition: Relationship between the item and the related resource, research output, or TerraLID entity (e.g., IsPartOf for a sample taken from an object). **Allowed values and other constraints:** controlled vocabulary

15.4 Type of resource

ID and name: B5.4 relation_resource
Provided by: data provider
Obligation: mandatory
Occurrences: 1–n
Definition: Type of resource or research output. Mandatory unless `B5.1.2 Type` = "TerraLID", as the type of a TerraLID entity follows from its TerraLID ID. **Allowed values and other constraints:** controlled vocabulary

15.5 Additional details

ID and name: B5.5 relation_detail
Provided by: data provider
Obligation: optional
Occurrences: 0–n
Definition: Additional information about the relation, e.g., if the related work addresses a specific aspect of the item (e.g., the geological setting).
Allowed values and other constraints: free text
Example: Information about the dating of the site.